Time Series Financial Fraud Detection

Sequential Transaction Analysis & Temporal Feature Engineering

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Agenda

1. Overview
2. Data & Features
3. Detection
4. Operations
5. Conclusions

Why Temporal Patterns Matter

Beyond Static Classification

Individual transaction features miss **behavioral shifts**:

- Burst attacks: 15 txns in 30 minutes
- Escalation: amounts growing 2x daily
- Impossible travel: NYC → Tokyo in 1 hour
- Night attacks: large txns at 3 AM

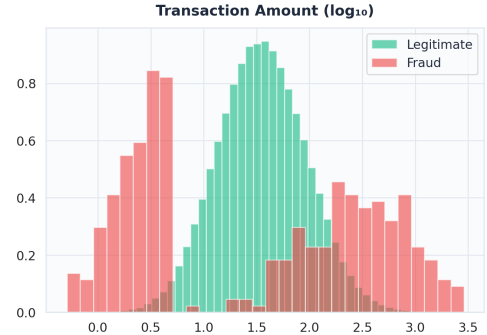
Dataset

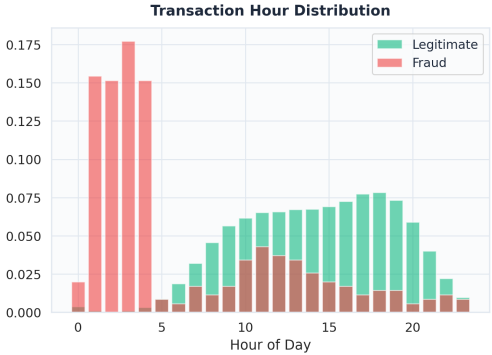
Customers	500
Period	180 days
Transactions	270K+
Fraud rate	∼0.1%
Fraud types	4

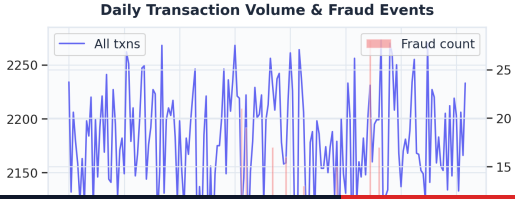
Our Approach

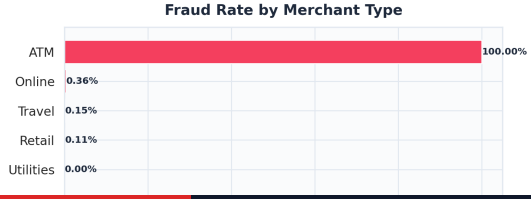
Engineer **temporal features** per-customer, then detect anomalies in the time series of spending behavior.

Transaction Overview

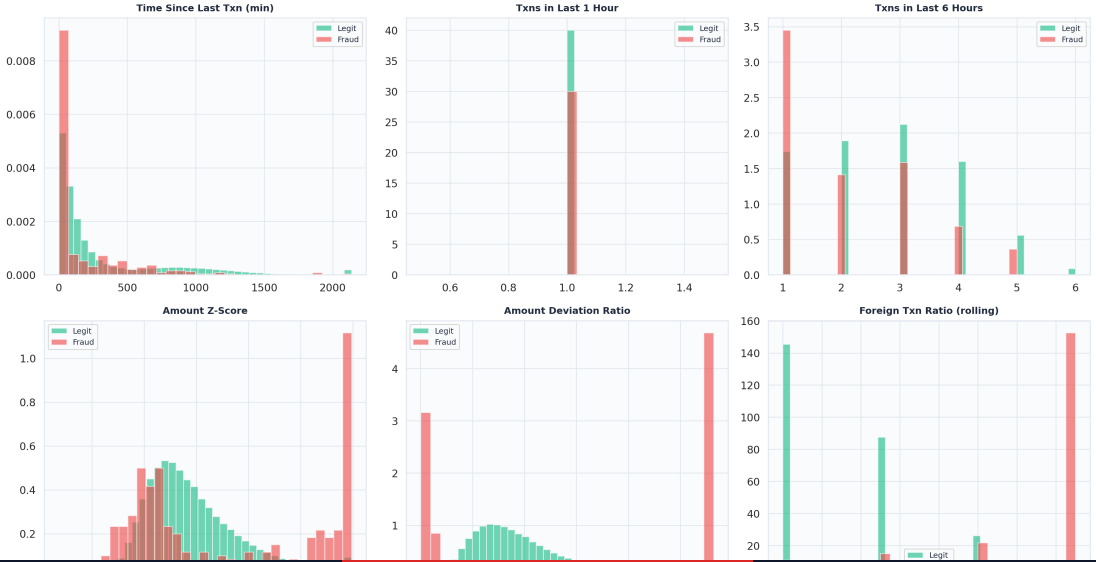






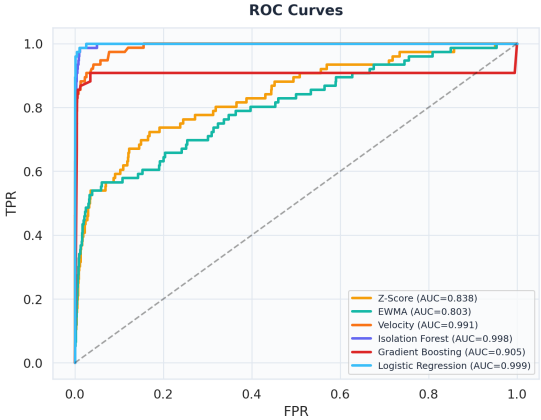


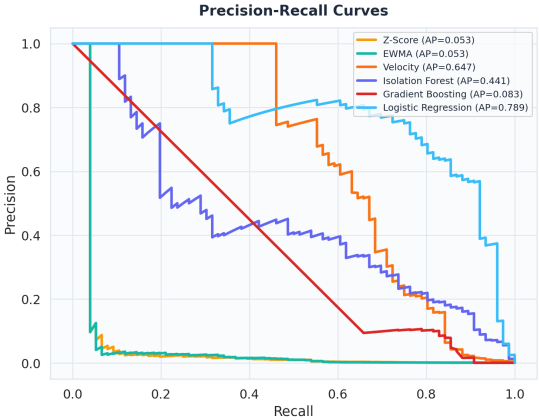
Temporal Feature Distributions: Legit vs Fraud



Fraud Pattern Examples

ROC & Precision-Recall Curves

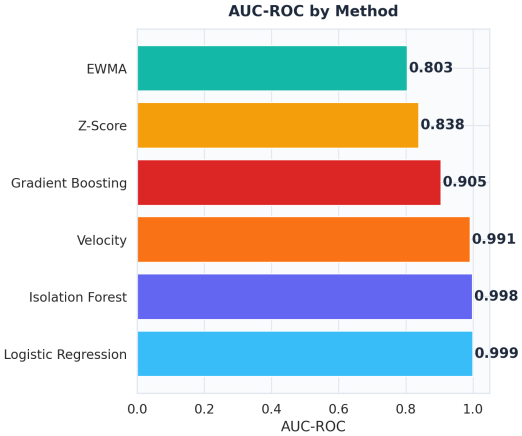


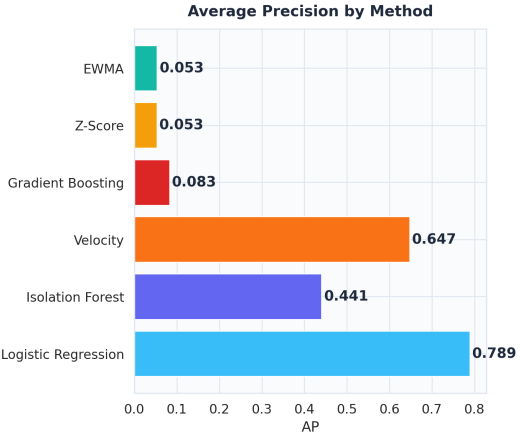


Result: AUC = 0.999

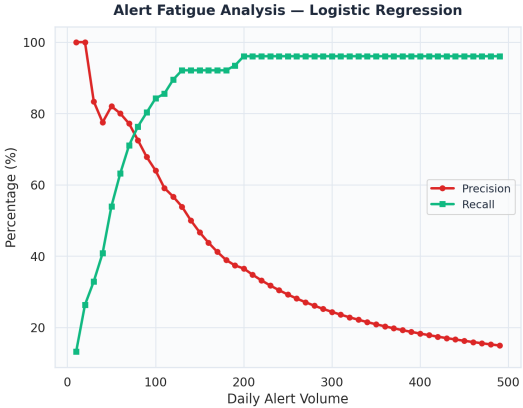
Logistic Regression with temporal features achieves near-perfect AUC — temporal context is the dominant signal.

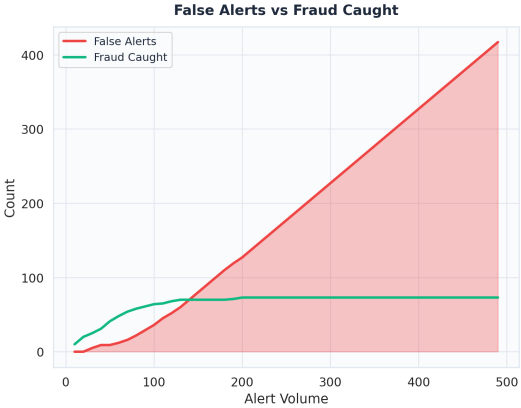
Method Comparison





Alert Fatigue Analysis

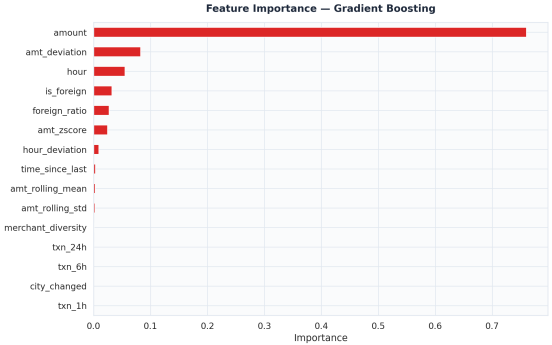


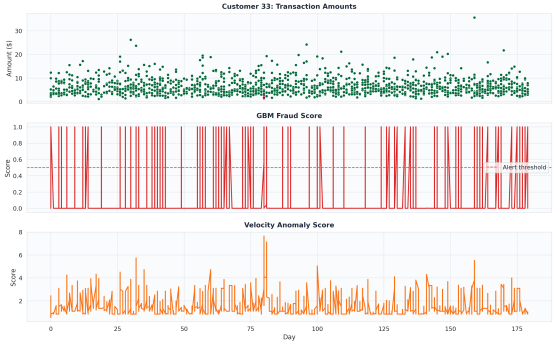


The Real Bottleneck

Precision degrades rapidly above 200 daily alerts. At 80% recall, 1 in 3 alerts is a false positive — setting the right threshold is an **operational** decision, not just a modeling one.

Feature Importance & Anomaly Timeline

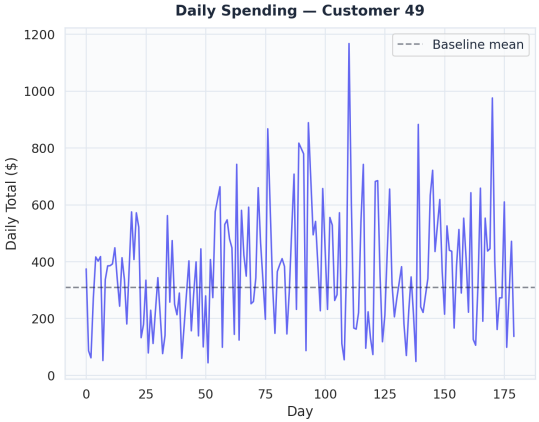


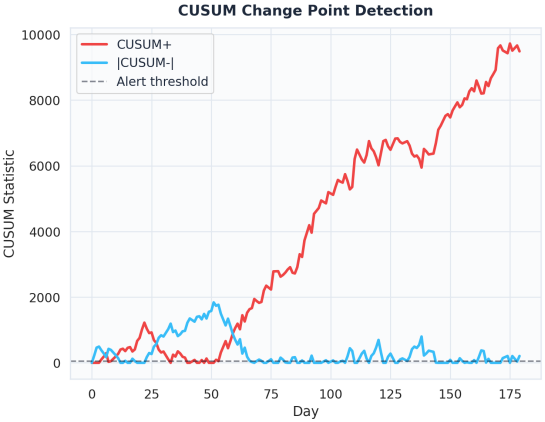


Top Features

Foreign ratio, amount deviation, and transaction velocity are the strongest temporal signals. The anomaly timeline shows scores spiking during fraud events.

Change Point Detection (CUSUM)

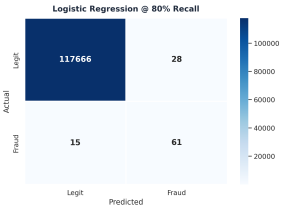




Regime Shift Detection

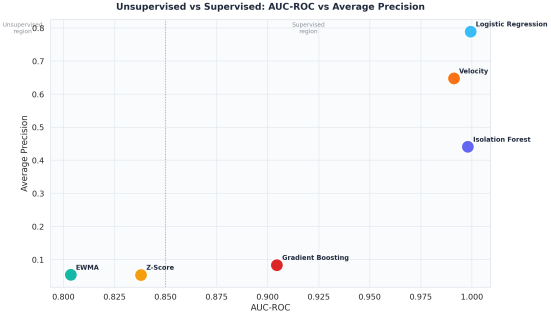
CUSUM detects when a customer's daily spending pattern shifts from its baseline — useful for catching slow-burn fraud like escalation patterns.

Operational Performance & Unsupervised vs Supervised



Operational Metrics

Metric	Value
Precision	0.6854
Recall	0.8026
F1-Score	0.7394
Threshold	0.9997
Alerts/day	89



Executive Summary

Time Series Financial Fraud Detection — Executive Summary

Fraud Detection Leaderboard

Method	AUC-ROC	Avg Precision	Type
Logistic Regression	0.9994	0.7890	Supervised
Isolation Forest	0.9981	0.4406	Unsupervised
Velocity	0.9913	0.6473	Unsupervised
Gradient Boosting	0.9045	0.0830	Supervised
Z-Score	0.8379	0.0532	Unsupervised
EWMA	0.8035	0.0535	Unsupervised

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FRAUD DETECTION SUMMARY

Transactions:  392,564
Customers:     500
Fraud rate:    0.089%
Period:        180 days

Best Model: Logistic Regression
  AUC-ROC:   0.9994
  Avg Prec:  0.7890

At 80% Recall:
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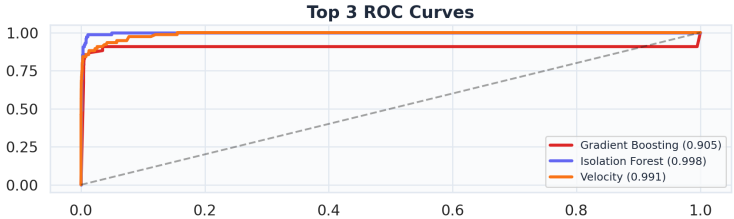

Production Architecture & Future Work

Production Pipeline

1. Real-time scoring (<50ms)
2. Tiered alerts: block / review / monitor
3. CUSUM on daily aggregates
4. Weekly model retraining
5. SHAP explanations for analysts

Future Work

1. LSTM/Transformer on sequences
2. Graph neural networks (fraud rings)
3. Federated learning across banks
4. Reinforcement learning for adaptive thresholds
5. Real-time streaming (Kafka + Flink)

Thank You

Questions & Discussion

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